

Problem 10.19

Calculate the effective annual interest rate equivalent to a nominal rate of discount of 6% compounded continuously.

Solution

With continuous compounding, the annual accumulation factor is

$$e^{\delta},$$

where the continuously compounded rate is

$$\delta = 0.06.$$

The equivalent effective annual interest rate i satisfies

$$1 + i = e^{\delta}.$$

Therefore,

$$\begin{aligned} i &= e^{0.06} - 1 \\ &\approx 0.06183654655. \end{aligned}$$

Hence, the equivalent effective annual interest rate is

$$\boxed{i \approx 6.183654655\%}.$$